

[HOME](#) [TOP](#) [CATALOG](#) [CONTESTS](#) [GYM](#) [PROBLEMSET](#) [GROUPS](#) [RATING](#) [EDU](#) [API](#) [CALENDAR](#) [HELP](#)

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[PROBLEMS](#) [SUBMIT CODE](#) [MY SUBMISSIONS](#) [STATUS](#) [STANDINGS](#) [CUSTOM INVOCATION](#)

A. Advising

time limit per test: 1 second

memory limit per test: 256 megabytes

In this problem, there are N courses in the curriculum and M requirements of the form "Course A has to be completed before course B ".

Your task is to find an order in which you can complete the courses. If there are multiple valid order, you may print any of them. If no such valid order exists, then print -1 .

Input

The first line contains an integer T ($1 \leq T \leq 10^5$), denoting the number of test cases.

For each test case, the first line of its input contains two integers N and M ($1 \leq N, M \leq 10^5$), denoting the number of courses and requirements. This is followed by M lines, each containing two integers A and B ($1 \leq A, B \leq N$) denoting that course A has to be completed before course B .

It is guaranteed that the sum of N and M over all test cases will not exceed $2 \cdot 10^5$.

Output

For each test case, print an order in which you can complete the courses. If there is no valid order, then print -1 .

Examples

input

Copy

```
1
6 6
4 2
5 6
1 4
2 3
6 3
1 6
```

output

Copy

```
1 5 4 6 2 3
```

input

Copy

```
3
1 1
1 1
6 5
4 3
1 3
3 6
6 4
2 3
4 2
3 1
4 1
```

output

Copy

CSE221 Assignment 06 Summer 2026

Contest is running

3 days

Contestant


[→ Submit?](#)

 Language: Java 21 64bit

Choose file:

Choose File

No file chosen



Success!



Submit



-1
-1
2 3 4 1

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The only programming contests Web 2.0 platform
Server time: Aug/20/2026 16:08:30^{UTC+6} (1).
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